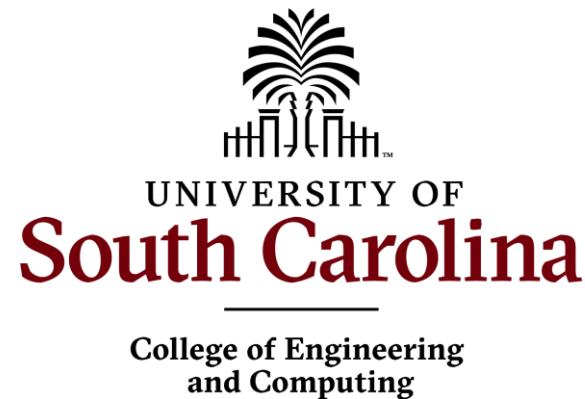


INTRODUCTION TO APPLIED NUMERICAL METHODS

Engineering Case Studies for Roots of Equations

Lang Yuan, Associate Professor



ANOTHER EXAMPLE

Use the Newton-Raphson method to locate the root of

$$f(x) = e^{-x} - x \quad \text{using an initial guess of } x_0 = 0$$

Solution:

The first derivative of $f(x)$ is

$$f'(x) = \frac{de^{-x}}{dx} - 1 = e^{-x} \frac{d(-x)}{dx} - 1 = -e^{-x} - 1$$



How to compute $\frac{de^{-x}}{dx}$?

Chain rule: $f(x) = F(g(x)) \longrightarrow \frac{df(x)}{dx} = \frac{df}{dg} \frac{dg}{dx}$

ANOTHER EXAMPLE

Use the Newton-Raphson method to locate the root of

$$f(x) = e^{-x} - x \quad \text{using an initial guess of } x_0 = 0$$

Solution:

The first derivative of $f(x)$ is $f'(x) = \frac{\partial e^{-x}}{\partial x} - 1 = e^{-x} \frac{\partial(-x)}{\partial x} - 1 = -e^{-x} - 1$

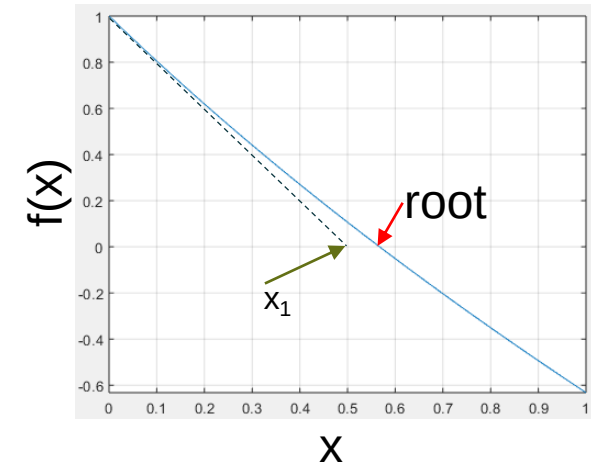
$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \quad \Rightarrow \quad x_{i+1} = x_i - \frac{e^{-x_i} - x_i}{-e^{-x_i} - 1}$$

Step 1: Substitute $x_0 = 0$ to obtain x_1

$$x_1 = 0 - \frac{e^0 - 0}{-e^0 - 1} = \frac{-1}{-2} = 0.5$$

Step 2: Substitute $x_1 = 0.5$ to obtain x_2

$$x_2 = 0.5 - \frac{e^{-0.5} - 0.5}{-e^{-0.5} - 1} = 0.5 - \frac{0.6065 - 0.5}{-0.6065 - 1} = 0.566311003$$

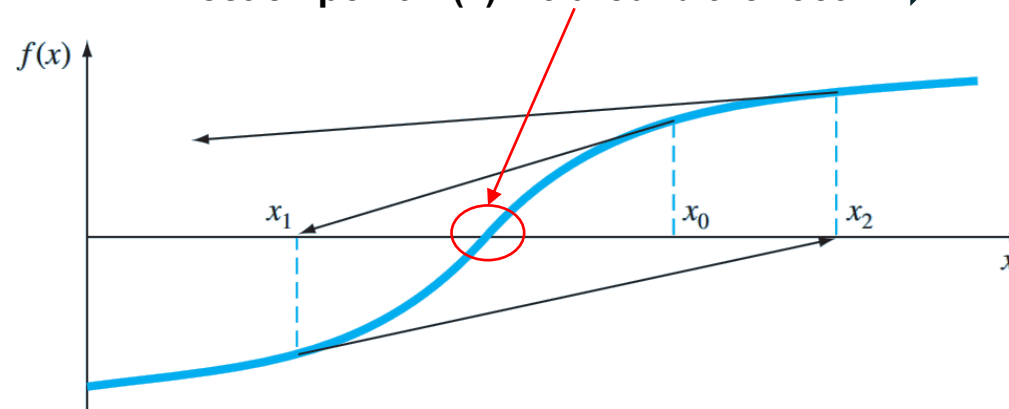


i	x_i	$\varepsilon_i(\%)$
0	0	100
1	0.500000000	11.8
2	0.566311003	0.147
3	0.567143165	0.0000220
4	0.567143290	$< 10^{-8}$

DIFFICULTIES WITH NEWTON-RAPHSON METHOD

Case 1:

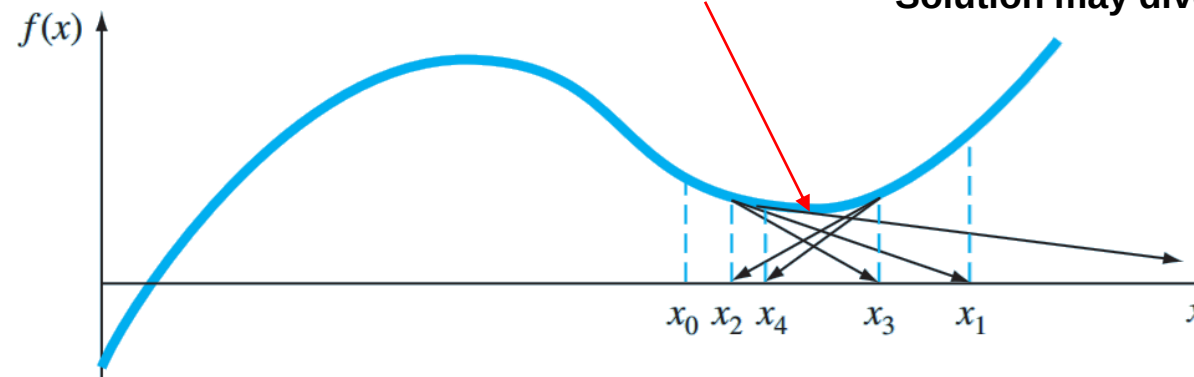
Inflection point $f''(x) = 0$ around the root $\Rightarrow$ Diverge from the root



Case 2:

A local minimum or maximum $\Rightarrow$

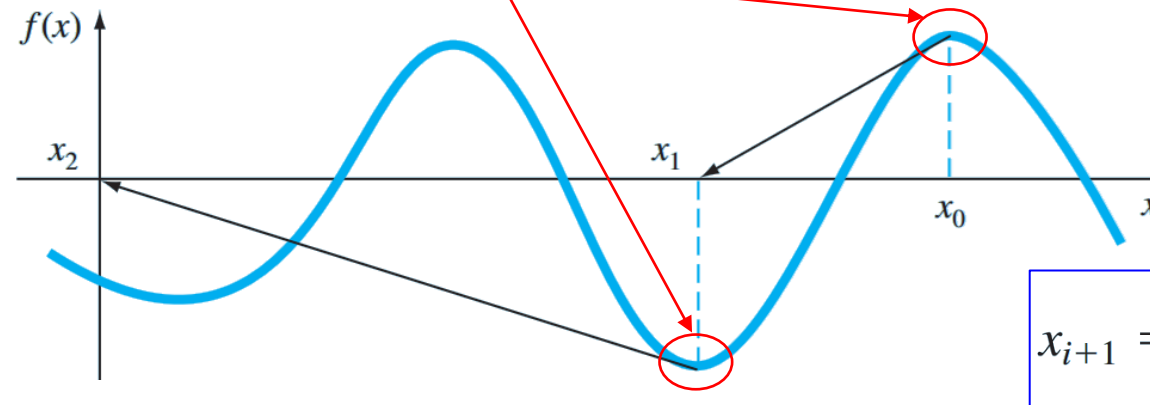
Oscillation around the minimum and maximum
Solution may diverge



DIFFICULTIES WITH NEWTON-RAPHSON METHOD

Case 3:

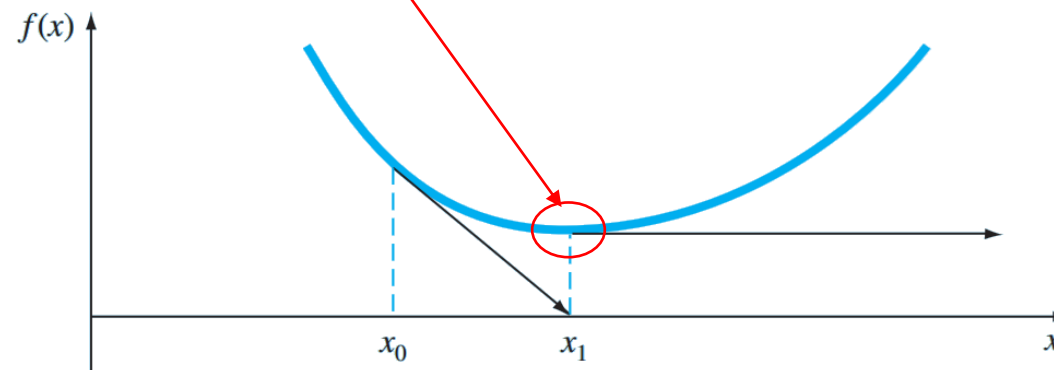
Near-zero slope $f'(x) = 0 \Rightarrow$ Jump away from the root (too large stride)



$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Case 4:

Near-zero slope $f'(x) = 0 \Rightarrow$ Jump away from the root (too large stride)



DIFFICULTIES WITH NEWTON-RAPHSON METHOD

There is no general guidance to ensure convergence for the Newton-Raphson method

Its convergence depends on

- The nature of the function
- The accuracy of the initial guess: initial guess close to the root as possible (my trick: use Matlab to plot the function first to get a good initial guess and then to check the solution)

Potential improvements

- Substitute the solution back into the original function to check the residual (in order to avoid case 2)
- Include an upper limit on the number of iterations (to avoid oscillating, slowly convergent, or divergent solutions)
- Check if $f'(x)$ approaches zero at any time during the computation

EXAMPLE: FLOW FRICTION FACTOR

Friction Factor for Turbulent Flow

$$\frac{1}{\sqrt{f}} = -2.0 \log_{10} \left(\frac{\varepsilon}{3.7D} + \frac{2.51}{\text{Re} \sqrt{f}} \right)$$

roughness
Colebrook Equation

Friction factor

Diameter

Reynolds number $\text{Re} = \frac{\rho V D}{\mu}$

Variables	Annotation	Value
f	Friction factor	To be solved [0.008 0.08]
ε	Roughness	0.0015 mm
D	Diameter	0.005 m
ρ	Density	1.23 kg/m ³
μ	Dynamic viscosity	1.79×10 ⁻⁵ N s/m ²
V	Velocity	40 m/s

Goal:

- Determine f for fluid flow through a smooth thin tube using parameters above given $e_s = 0.01\%$

EXAMPLE: FLOW FRICTION FACTOR

Use the Bisection method to locate the root of

$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log_{10} \left(8.1081^{-5} + \frac{1.8264^{-4}}{\sqrt{f}} \right) \text{ within the interval } [0.008 \ 0.08]$$

Solution:

Step 1: Evaluate the bounds

$$g(0.008) = 5.834$$

$$g(0.08) = -2.7416$$

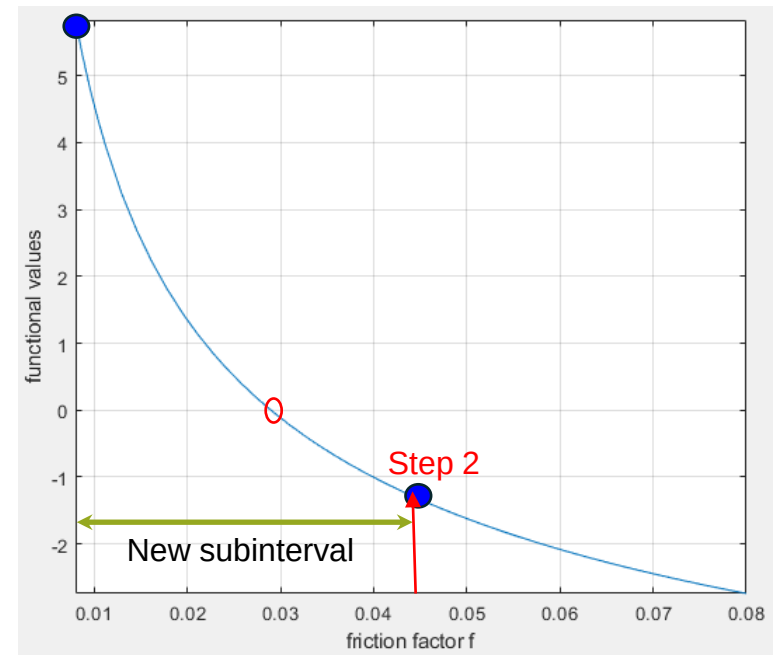
Step 2: Estimate the root x_r by setting

$$x_r = \frac{0.008 + 0.08}{2} = 0.044$$

$$g(0.044) = -1.2756$$

$$f_u = f_r = 0.044$$

$$\varepsilon_a = \left| \frac{f_{new} - f_{old}}{f_{new}} \right| = \left| \frac{0.044 - 0.08}{0.044} \right| = 81.81\%$$



EXAMPLE: FLOW FRICTION FACTOR

Use the Bisection method to locate the root of

$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log_{10} \left(8.1081^{-5} + \frac{1.8264^{-4}}{\sqrt{f}} \right) \text{ with the interval } [0.008 \ 0.044]$$

Solution:

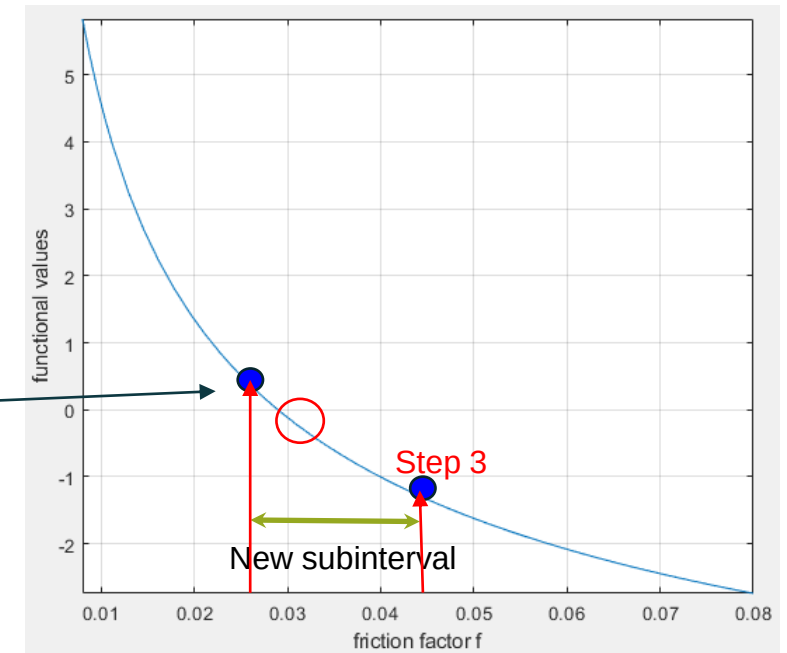
Step 3: Estimate the root x_r by setting

$$f_r = \frac{f_l + f_u}{2} = \frac{0.008 + 0.044}{2} = 0.026$$

$$g(0.026) = 0.37 \quad \Rightarrow \quad \varepsilon_a = \left| \frac{f_{\text{new}} - f_{\text{old}}}{f_{\text{new}}} \right| = \left| \frac{0.026 - 0.044}{0.026} \right| = 69.23\%$$

$$g(l)g(r) = g(0.008)g(0.026) = 2.159 > 0$$

$$f_l = f_r = 0.026 \quad g(l) = g(r) = 0.37$$



EXAMPLE: FLOW FRICTION FACTOR

Use the Bisection method to locate the root of

$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log_{10} \left(8.1081^{-5} + \frac{1.8264^{-4}}{\sqrt{f}} \right) \text{ within the interval } [0.028953 \ 0.02899]$$

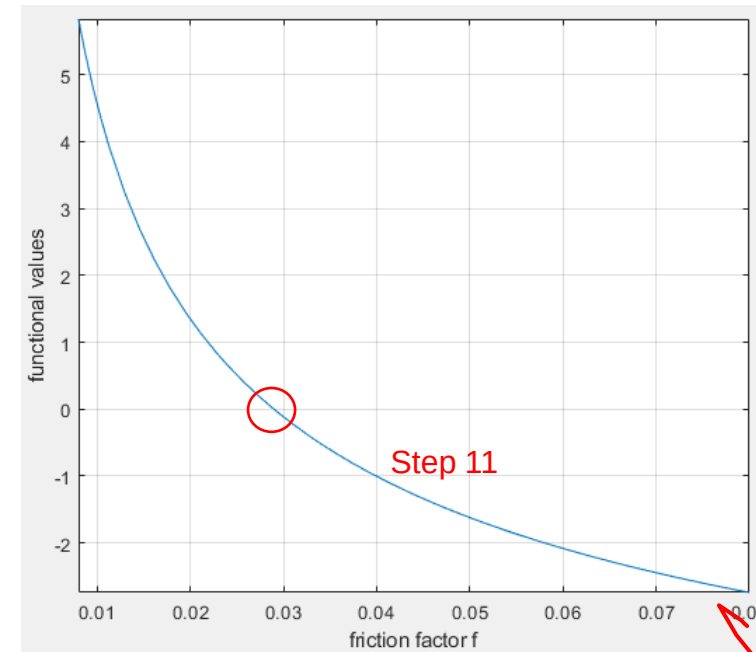
Solution:

Step 11: Estimate the root x_r by setting

$$f_r = \frac{f_l + f_u}{2} = \frac{0.028953 + 0.02899}{2} = 0.02897$$

$$g(0.02897) = -3.3686 \times 10^{-4}$$

$$\varepsilon_a = \left| \frac{f_{\text{new}} - f_{\text{old}}}{f_{\text{new}}} \right| = \left| \frac{0.02897 - 0.02899}{0.02897} \right| = 0.06\%$$



EXAMPLE: FLOW FRICTION FACTOR

$$\frac{d}{dx} \ln(x) = \frac{1}{x}, \quad \frac{d}{dx} \log_a(x) = \frac{1}{x \ln(a)}.$$

Use the Newton-Raphson method to locate the root of

$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log_{10} \left(8.1081^{-5} + \frac{1.8264^{-4}}{\sqrt{f}} \right) \quad \text{With starting point of 0.008}$$

$$g(h) = h + 2.0 \log_{10}(a + bh), \text{ given } h = \frac{1}{\sqrt{(f)}}, a = 8.1081^{-5}, b = 1.8264^{-4}$$

$$g'(h) = 1 + \frac{2.0b}{(a + bh)\ln(10)}$$

$$h_{i+1} = h_i - \left(\frac{h_i + 2.0 \log_{10}(a + bh_i)}{1 + \frac{2.0b}{(a + bh_i)\ln(10)}} \right)$$

EXAMPLE: FLOW FRICTION FACTOR

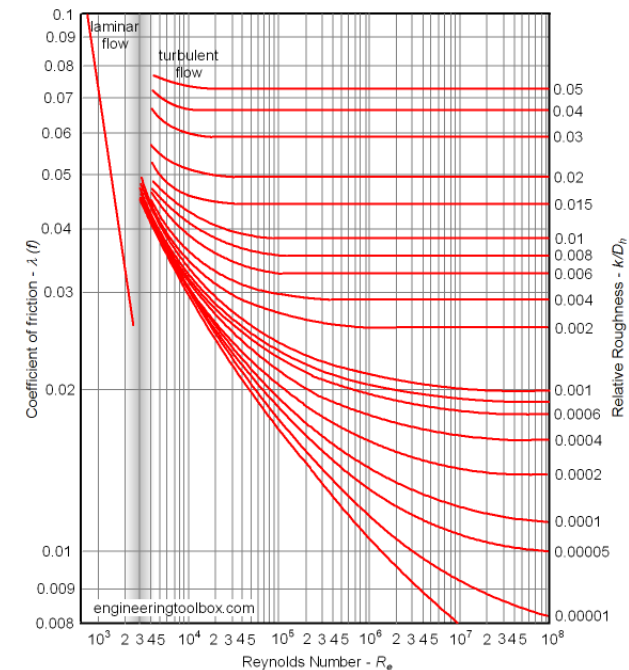
- Step 1, $h_i = \frac{1}{\sqrt{0.008}} = 11.18, h_{i+1} = 5.752, \varepsilon_a = \left| \frac{h_{i+1} - h_i}{h_{i+1}} \right| = 0.94$
- Step 2, $h_i = 5.7517, h_{i+1} = 5.8753, \varepsilon_a = \left| \frac{h_{i+1} - h_i}{h_{i+1}} \right| = 0.02$
- Step 3, $h_i = 5.8753, h_{i+1} = 5.8755, \varepsilon_a = \left| \frac{h_{i+1} - h_i}{h_{i+1}} \right| = 2.55e - 5$
- $f = \left(\frac{1}{h} \right)^2 = 0.02897$

EXAMPLE: FLOW FRICTION FACTOR

Technical Findings

- The bisection can solve the problem robustly
- Newton-Raphson method can potentially solve the problem more efficiently
 - Takes 6 iterations (as opposed to 12 iterations for the bisection)
 - A poor initial guess causes divergence of the program However the derivative of the $g(f)$ may be a little involved
- The numerical solution process can be repeated to generate friction factor graph, which can be used as a look-up table

$$\frac{1}{\sqrt{f}} = -2.0 \log_{10} \left(\frac{\varepsilon}{3.7D} + \frac{2.51}{\text{Re} \sqrt{f}} \right)$$



$$g = g(\varepsilon/D, \text{Re})$$